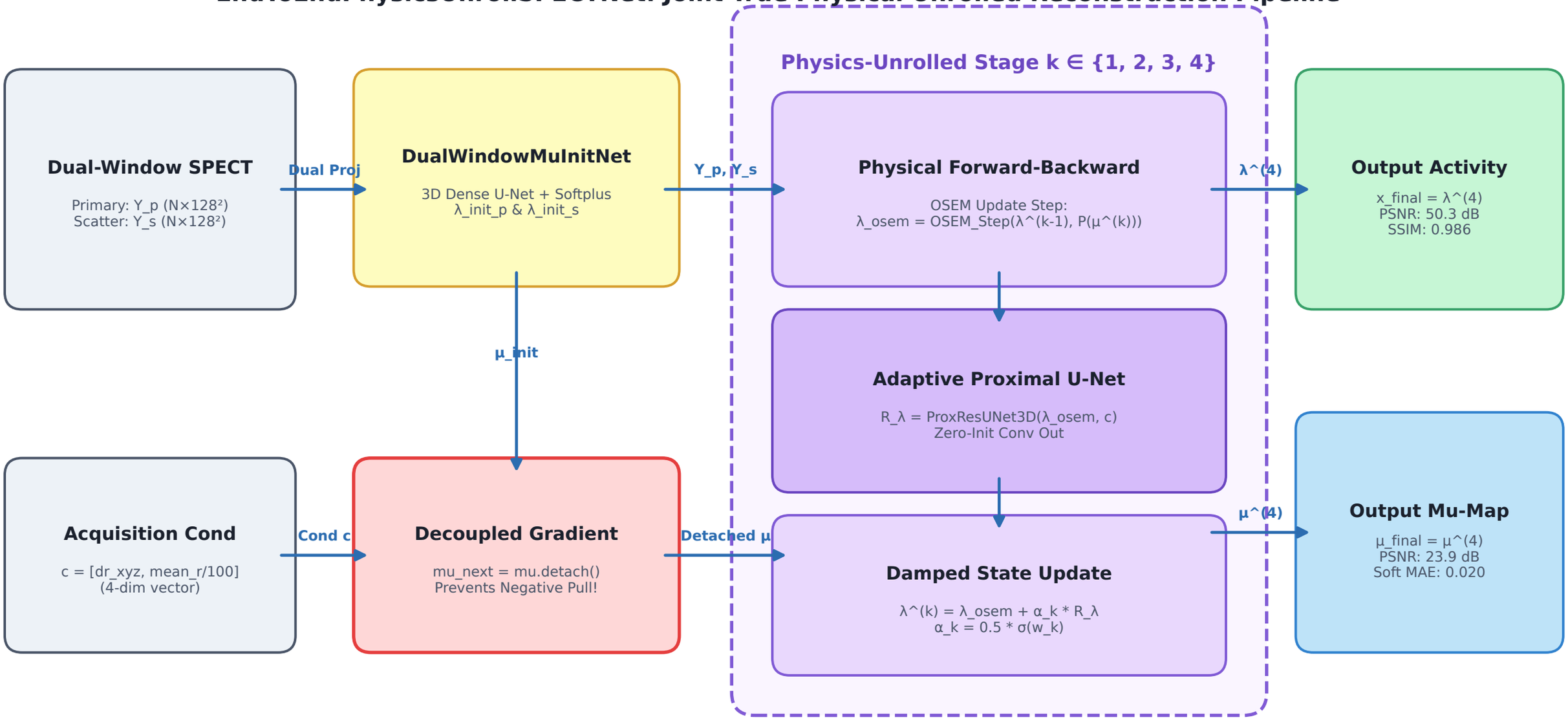
EndToEndPhysicsUnrollSPECTNet: Joint True Physical Unrolled Reconstruction Pipeline
Dual-Window SPECT
Primary: Y_p (N×128²)
Scatter: Y_s (N×128²)
Dual Proj
DualWindowMuInitNet
3D Dense U-Net + Softplus
λ_init_p & λ_init_s
Y_p, Y_s
Physics-Unrolled Stage k ∈ {1, 2, 3, 4}
Physical Forward-Backward
OSEM Update Step:
λ_osem = OSEM_Step(λ^(k-1), P(μ^(k)))
λ^(4)
Output Activity
x_final = λ^(4)
PSNR: 50.3 dB
SSIM: 0.986
μ_init
Adaptive Proximal U-Net
R_λ = ProxResUNet3D(λ_osem, c)
Zero-Init Conv Out
Acquisition Cond
c = [dr_xyz, mean_r/100]
(4-dim vector)
Cond c
Decoupled Gradient
mu_next = mu.detach()
Prevents Negative Pull!
Detached μ
Damped State Update
λ^(k) = λ_osem + α_k * R_λ
α_k = 0.5 * σ(w_k)
μ^(4)
Output Mu-Map
μ_final = μ^(4)
PSNR: 23.9 dB
Soft MAE: 0.020